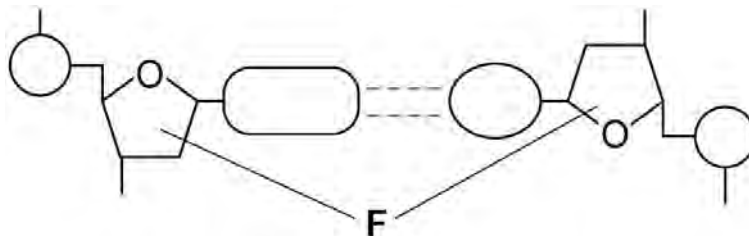


4 **Figure 4** shows one base pair of a DNA molecule.

Figure 4



0 4 . 1 Name part **F** of each nucleotide.

[1 mark]

0 4 . 2 Scientists determined that a sample of DNA contained 18% adenine.

What were the percentages of thymine and guanine in this sample of DNA?

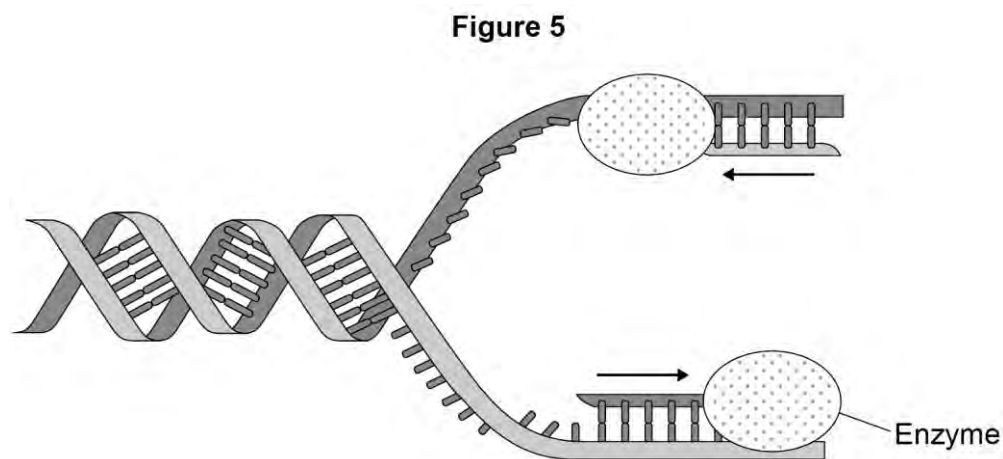
[2 marks]

Percentage of thymine

Percentage of guanine

During replication, the two strands of a DNA molecule separate and each acts as a template for the production of a new strand.

Figure 5 represents DNA replication.



0 4 . 3 Name the enzyme shown in **Figure 5**.

[1 mark]

The arrows in **Figure 5** show the directions in which each new DNA strand is being produced.

0 4 . 4 Use **Figure 4**, **Figure 5** and your knowledge of enzyme action to explain why the arrows point in opposite directions.

[4 marks]
